Notes: Ostrovsky (AER, 2008)

Stability in Supply Chain Networks

Contents

1 Big picture

- **Question.** How can the theory of stable matching be extended from two-sided markets to vertically connected supply chains with suppliers, intermediaries, and final buyers?
- **Setting.** Agents are nodes in a directed upstream–downstream network. Contracts specify a seller, a buyer, a unit identifier, and possibly a price.
- **Core object.** A network is a set of bilateral contracts. It describes who trades with whom, in which direction, for which unit, and at what price.
- **Main solution concept.** A network is **chain stable** if it is individually rational and no upstream–downstream chain of contracts can block it.
- **Main preference restrictions.** Agents' preferences must satisfy **same-side substitutability** and **cross-side complementarity**.
- **Same-side substitutability.** If more contracts become available on one side of an agent, the agent does not newly accept contracts on that same side that it previously rejected.
- **Cross-side complementarity.** If more contracts become available on one side, the agent becomes more willing to accept contracts on the opposite side. More buyers make inputs more valuable; more inputs make buyers more valuable.
- **Main theorem.** Under these restrictions, a chain-stable network exists.
- **Algorithmic contribution.** The proof constructs a fixed-point algorithm, the T-algorithm, which generalizes deferred acceptance and salary adjustment algorithms.
- **Side-optimal networks.** The algorithm starting from the smallest pre-network finds the supplier-optimal stable network. Starting from the largest pre-network finds the consumer-optimal stable network.
- **Comparative statics.** Adding a supplier makes other suppliers worse off and final consumers better off at the side-optimal stable networks. Adding a final consumer has the opposite effect.
- **Broader contribution.** The paper provides a matching-theoretic foundation for vertical supply chains, market design with more than two types of agents, and empirical restrictions for supply-chain data.

2 Model primitives and measurement

2.1 Agents and upstream–downstream order

- The industry contains a finite set of agents or nodes A.
- A partial order $\succ$ fixes possible trading directions. If $a \succ b$, then b is downstream from a, so a can potentially sell to b.
- The partial order is acyclic: an agent cannot be its own direct or indirect supplier.
- Nodes can be suppliers of basic inputs, intermediaries, or consumers of final outputs.

Interpretation: The paper generalizes two-sided matching by allowing intermediate nodes that both buy and sell. The acyclicity assumption preserves a supply-chain direction.

2.2 Contracts

A contract is a vector

$$c = (s, b, \ell, p), \tag{1}$$

where s is the seller, b is the buyer, ℓ is the serial number of the unit, and p is the price paid by the buyer to the seller.

- Multiple contracts between the same seller and buyer can represent multiple units or different products.
- The finite set of possible contracts is denoted C.
- Restrictions such as embargoes or technological impossibilities can be imposed by deleting contracts from C.

Interpretation: The contract notation is flexible enough to include wages, prices, identities, quantities, and other contract characteristics.

2.3 Preferences and choice

For agent a and set of contracts X, let $Ch_a(X)$ denote a's most preferred subset of X. Let $U_a(X)$ be the contracts in which a is the buyer, and $D_a(X)$ be the contracts in which a is the seller.

In the quasi-linear benchmark,

$$V_a(X) = W_a(\{(s_c, b_c, \ell_c) : c \in X\}) + \sum_{c \in D} p_c - \sum_{c \in U} p_c. \tag{2}$$

Interpretation: For upstream suppliers, W_a captures production costs. For final consumers, it captures consumption value. For intermediaries, it captures the cost of converting inputs into outputs.

node can be involved in several contracts, some as a seller, some as a buyer, bu
ıvolved in two contracts that differ only in price p, i.e., it cannot buy or sell tl
ce. Nodes have preferences over sets of contracts that involve them as the buye
or example, in the simplest case of quasi-linear utilities and profits, the utility of
l in a set of contracts X is

$$V_a(X) = W_a(\{(s_{\mathbf{c}}, b_{\mathbf{c}}, l_{\mathbf{c}}) | \mathbf{c} \in X\}) + \sum_{\mathbf{c} \in D} p_{\mathbf{c}} - \sum_{\mathbf{c} \in U} p_{\mathbf{c}},$$

$= \{\mathbf{c} \in X | a = s_{\mathbf{c}}\}$ and $U = \{\mathbf{c} \in X | a = b_{\mathbf{c}}\}$, i.e., D is the set of contracts in X iı

2.4 Same-side substitutability

Preferences are same-side substitutable if increasing the set of available contracts on one side does not cause the agent to newly accept contracts on that same side that it previously rejected.

Interpretation: Inputs are substitutes for other inputs, and outputs are substitutes for other outputs. This rules out strong economies of scale and many complementary-input technologies.

2.5 Cross-side complementarity

Preferences are cross-side complementary if increasing the set of contracts on one side makes the agent weakly more willing to accept contracts on the other side.

Interpretation: Inputs and outputs are complements. More available buyers make inputs more valuable; more available inputs make downstream sales more valuable.

Preferences of agent a are *same-side substitutable* if for any two sets of contracts X and Y such that $D(X) = D(Y)$ and $U(X) \subset U(Y)$, $U(X) \setminus U(Ch(X)) \subset U(Y) \setminus U(Ch(Y))$, and for any two sets X and Y such that $U(X) = U(Y)$ and $D(X) \subset D(Y)$, $D(X) \setminus D(Ch(X)) \subset D(Y) \setminus D(Ch(Y))$. That is, preferences are same-side substitutable if, choosing from a bigger set of contracts on one side, the agent does not accept any contracts *on that side* that he rejected when he was choosing from the smaller set.

Preferences of agent a are *cross-side complementary* if for any two sets of contracts X and Y such that $D(X) = D(Y)$ and $U(X) \subset U(Y)$, $D(Ch(X)) \subset D(Ch(Y))$, and for any two sets X and Y such that $D(X) \subset D(Y)$ and $U(X) = U(Y)$, $U(Ch(X)) \subset U(Ch(Y))$. That is, preferences are cross-side complementary if, when presented with a bigger set of contracts on one side, an agent does not reject any contract *on the other side* that he accepted before.

Same-side substitutability is a generalization of the gross substitutes condition introduced by Kelso and Crawford (1982) and used widely in the matching literature. If there are only two sides

2.6 Networks and individual rationality

A network μ is a set of contracts. Let $\mu(a)$ denote the set of contracts involving agent a. The network is individually rational if

$$Ch_a(\mu(a)) = \mu(a) \quad \text{for all } a \in A. \tag{3}$$

Interpretation: No agent wants to unilaterally drop any of its existing contracts.

2.7 Chain blocks and chain stability

A chain is a sequence of contracts $\{c_1, \ldots, c_n\}$ such that the buyer in c_i is the seller in c_{i+1}. A chain block of network μ is a chain not already in μ such that every agent along the chain would be willing to add its new contract(s), possibly after dropping some existing contracts. A network is **chain stable** if it is individually rational and has no chain blocks.

Interpretation: Chain stability is the supply-chain analogue of pairwise stability. In a two-sided market, a blocking chain has length one and the concept reduces to pairwise stability.

B. *Stable Networks*

In the model, relationships between the nodes are represented by contracts. I call a collection of relationships between the nodes in a market a *network*, i.e., a network is simply a set of contracts. Let $\mu(a)$ denote the set of contracts involving a in network μ. Network μ is *individually rational* if, for any agent a, $Ch_a(\mu(a)) = \mu(a)$, i.e., no agent would like to unilaterally drop any of his contracts.

The most widely used solution concept in the two-sided matching literature is pairwise stability. Its analogue in the supply chain setting is chain stability, defined as follows. A *chain* is a sequence of contracts, $\{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$, $n \geq 1$, such that for any $i < n$, $b_{\mathbf{c}_i} = s_{\mathbf{c}_{i+1}}$, i.e., the buyer in contract $\mathbf{c}_i$ is the same node as the seller in contract $\mathbf{c}_{i+1}$. Note that the chain does not have to go all the way from one of the most upstream nodes in the market to one of the most downstream nodes; it can connect several nodes in the middle of the market. For notational convenience, let $b_i \equiv b_{\mathbf{c}_i}$ and $s_i \equiv s_{\mathbf{c}_i}$. For a network μ, a *chain block* is a chain $\{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ such that

- $\forall i \leq n$, $\mathbf{c}_i \notin \mu$,
- $\mathbf{c}_1 \in Ch_{s_1}(\mu(s_1) \cup \mathbf{c}_1)$,

3 Models and empirical specifications

3.1 Pre-networks and arrows

A pre-network is a set of directed arrows. Each arrow is

$$r = (o_r, d_r, c_r), \tag{4}$$

where o_r is the origin, d_r is the destination, and c_r is the attached contract.

- If the origin is the seller of c_r, the arrow is downstream.
- If the origin is the buyer of c_r, the arrow is upstream.
- For a pre-network ν, $\nu(a)$ is the set of contracts attached to arrows pointing to a.

Interpretation: Arrows represent unilateral willingness to form contracts. A contract belongs to the final network only if both sides point to it.

3.2 The T mapping

For any pre-network ν, define

$$T(\nu) = \{r \in R : c_r \in Ch_{o_r}(\nu(o_r) \cup c_r)\}. \tag{5}$$

Interpretation: In $T(\nu)$, each node points to the contracts it would choose if it could also choose from all contracts currently pointing to it.

3.3 The F mapping

Given a pre-network ν, define

$$F(\nu) = \{c \in C : (s_c, b_c, c) \in \nu \text{ and } (b_c, s_c, c) \in \nu\}. \tag{6}$$

Interpretation: F keeps only mutually acceptable contracts, i.e., contracts with arrows in both directions.

3.4 Fixed points and stable networks

Lemma 1. If $T(\nu^*) = \nu^*$, then $\mu^* = F(\nu^*)$ is chain stable. Conversely, each chain-stable network corresponds to exactly one fixed-point pre-network.

Interpretation: The core mathematical strategy is to prove existence of a fixed point of T, then map it into a stable network using F.

> Mapping T on the set of pre-networks is defined as follows. Consider any pre-network ν. Then pre-network $T(\nu)$ consists of such arrows $\mathbf{r}$ that the contract attached to the arrow, $\mathbf{c_r}$, belongs to the most preferred subset chosen by the arrow's origin, $o_\mathbf{r}$, from set $(\nu(o_\mathbf{r}) \cup \mathbf{c_r})$, i.e.,
>
> $$T(\nu) = \{\mathbf{r} \in R | \mathbf{c_r} \in Ch_{o_\mathbf{r}}(\nu(o_\mathbf{r}) \cup \mathbf{c_r})\}.$$
>
> Pre-network ν^* is a *fixed point* of mapping T if $T(\nu^*) = \nu^*$.
>
> The first step of the proof is Lemma 1 below. It shows that fixed points of mapping T play a very special role: there is a one-to-one correspondence between the set of chain-stable networks and the set of fixed points of mapping T. Thus, to prove the main result, it will be sufficient to construct a fixed-point pre-network.
>
> Formally, define mapping F from the set of pre-networks to the set of networks as follows. Take any pre-network ν, and consider any contract $\mathbf{c}$. Contract $\mathbf{c}$ belongs to $\mu = F(\nu)$ if, and only if, both the arrow from the seller of $\mathbf{c}$ to the buyer of $\mathbf{c}$ with $\mathbf{c}$ attached and the arrow from the buyer of $\mathbf{c}$ to the seller of $\mathbf{c}$ with $\mathbf{c}$ attached are contained in ν, i.e.,
>
> $$F(\nu) = \{\mathbf{c} \in C | (s_\mathbf{c}, b_\mathbf{c}, \mathbf{c}) \in \nu \text{ and } (b_\mathbf{c}, s_\mathbf{c}, \mathbf{c}) \in \nu\}.$$
>
> LEMMA 1: *For any pre-network ν^* such that $T(\nu^*) = \nu^*$, network $\mu^* = F(\nu^*)$ is chain stable. Moreover, for any chain-stable network μ^*, there exists exactly one fixed-point pre-network ν^* such that $\mu^* = F(\nu^*)$.*

3.5 Ordering pre-networks

For pre-networks ν_1 and ν_2, define $\nu_1 \leq \nu_2$ if

- downstream arrows in ν_1 are a subset of downstream arrows in ν_2;
- upstream arrows in ν_1 are a superset of upstream arrows in ν_2.

Let $\nu_{\min}$ contain all upstream arrows and no downstream arrows. Let $\nu_{\max}$ contain all downstream arrows and no upstream arrows.

3.6 The T-algorithm

1. Start from $\nu_{\min}$.
2. Iterate T: $\nu_{\min}, T(\nu_{\min}), T^2(\nu_{\min}), \ldots$.
3. Since the set of pre-networks is finite and T is isotone, the sequence reaches a fixed point $\nu^*_{\min}$.
4. Apply F to get $\mu^*_{\min} = F(\nu^*_{\min})$.

Theorem 1. There exists a chain-stable network.

Interpretation: The algorithm is a many-layer analogue of deferred acceptance. It starts from the extreme upstream-proposing direction and converges to a stable network.

To prove that chain-stable networks exist, it is now sufficient to show that mapping T has a fixed point. To find a fixed point, I first introduce a partial ordering on the set of pre-networks, as follows. Let ν_1 and ν_2 be two pre-networks. Then, ν_1 is said to be less than or equal to ν_2 ($\nu_1 \leq \nu_2$) if the set of downstream arrows in ν_1 is a *subset* of the set of downstream arrows in ν_2, while the set of upstream arrows in ν_1 is a *superset* of the set of upstream arrows in ν_2. Let $\nu_{\min}$ be the pre-network that includes all possible upstream arrows and no downstream arrows, and let $\nu_{\max}$ be the pre-network that includes no upstream arrows and all possible downstream arrows. By construction, for any pre-network ν, $\nu_{\min} \leq \nu \leq \nu_{\max}$.

The following key lemma shows that mapping T is isotone, i.e., order-preserving.

LEMMA 2: *For any pair of pre-networks ν_1 and ν_2 such that $\nu_1 \leq \nu_2$, we have $T(\nu_1) \leq T(\nu_2)$.*

Intuitively, if $\nu_1 \leq \nu_2$, then in ν_1 each node a has fewer "options" (i.e., contracts attached to arrows pointing to it) from potential sellers and more "options" from potential buyers than in ν_2. Then, by same-side substitutability and cross-side complementarity, given its options in ν_1, node a will be less "picky" in forming contracts with potential sellers and more "picky" in forming contracts with potential buyers than it would be given its options in ν_2. Hence, for any node a there will be more arrows pointing upstream from a (and fewer arrows pointing downstream) in $T(\nu_1)$ than in $T(\nu_2)$.

It is now easy to construct an algorithm for finding a chain-stable network. Namely, take the smallest pre-network, $\nu_{\min}$. Apply mapping T to it, getting $T(\nu_{\min})$. Since $\nu_{\min}$ is the smallest pre-network, by definition, $\nu_{\min} \leq T(\nu_{\min})$. Now, apply mapping T to $T(\nu_{\min})$. By Lemma 2, $T(\nu_{\min}) \leq T(T(\nu_{\min})) = T^2(\nu_{\min})$. Applying mapping T repeatedly, we get an increasing sequence of pre-networks: $\{\nu_{\min}, T(\nu_{\min}), T^2(\nu_{\min}), T^3(\nu_{\min}), \ldots\}$. Since the set of all pre-networks is finite, after a finite number of steps this sequence has to converge to a fixed point $\nu^*_{\min}$. We can now apply mapping F to this fixed-point pre-network to get a chain-stable network $\mu^*_{\min}$. This completes the description of the algorithm, and also proves Theorem 1, stated below.

THEOREM 1: *There exists a chain-stable network.*

4 Identification and theoretical design

4.1 Why the preference restrictions are enough

- Same-side substitutability makes T order preserving on the same side of trade.

- Cross-side complementarity makes T order preserving across the upstream and downstream sides.
- Together they make T isotone, allowing a fixed-point argument.

Interpretation: The existence theorem is not merely a compactness result. It depends on a monotonicity structure that comes from the economics of substitutes on one side and complements across sides.

4.2 Side-optimal stable networks

Starting from $\nu_{\min}$ produces the supplier-optimal stable network $\mu^*_{\min}$. Starting from $\nu_{\max}$ produces the consumer-optimal stable network $\mu^*_{\max}$.

Theorem 2. Suppliers of basic inputs weakly prefer $\mu^*_{\min}$ to any other chain-stable network and weakly prefer any chain-stable network to $\mu^*_{\max}$. Consumers of final outputs have the opposite ranking.

> stable network, and all consumers of final outputs are at least as well off in network $\mu_{\max}$ as they are in any other chain-stable network.
>
> THEOREM 2: *Let $\mu^*_{\min} = F(\nu^*_{\min})$, $\mu^*_{\max} = F(\nu^*_{\max})$, and let μ^* be a chain-stable network. Then any $a \in \overline{A}$ (weakly) prefers $\mu^*_{\min}$ to μ^* and μ^* to $\mu^*_{\max}$, and any $a \in \underline{A}$ (weakly) prefers $\mu^*_{\max}$ to μ^* and μ^* to $\mu^*_{\min}$.*

Interpretation: Side-optimality generalizes the man-optimal and woman-optimal stable matchings in Gale-Shapley matching.

4.3 Comparative statics

Theorem 3. Adding a new supplier of basic inputs makes other suppliers weakly worse off and final consumers weakly better off at the side-optimal stable networks. Adding a final consumer has the symmetric effect.

> stable matchings in A'.
>
> THEOREM 3: *If $U_{a'}(A) = \emptyset$, i.e., a' is a supplier of basic inputs, then each $a \in \overline{A}$ is* at least *as well off in $\mu^*_{\max}$ as in $\mu'_{\max}$ and* at least *as well off in $\mu^*_{\min}$ as in $\mu'_{\min}$; each $a \in \underline{A}$ is* at most *as well off in $\mu^*_{\max}$ as in $\mu'_{\max}$ and* at most *as well off in $\mu^*_{\min}$ as in $\mu'_{\min}$.*
>
> *Symmetrically, if $D_{a'}(A) = \emptyset$, i.e., a' is a consumer of final outputs, then each $a \in \overline{A}$ is* at most *as well off in $\mu^*_{\max}$ as in $\mu'_{\max}$ and* at most *as well off in $\mu^*_{\min}$ as in $\mu'_{\min}$; each $a \in \underline{A}$ is* at least *as well off in $\mu^*_{\max}$ as in $\mu'_{\max}$ and* at least *as well off in $\mu^*_{\min}$ as in $\mu'_{\min}$.*

Interpretation: More competition on one side benefits the opposite side and hurts the same side, extending classic matching-market comparative statics.

4.4 Alternative solution concepts

Theorem 4. Under the same preference restrictions, the set of tree-stable networks equals the set of chain-stable networks.

Theorem 5. If each node can have at most one upstream contract and at most one downstream contract, the set of chain-stable networks equals the weak core.

not blocked by any tree.

THEOREM 4: *Under same-side substitutability and cross-side complementarity, the set of tree-stable networks is equal to the set of chain-stable networks.*

THEOREM 5: *If each node $a \in A$ can have at most one upstream contract and at most one downstream contract and has same-side substitutable and cross-side complementary preferences, then the set of chain-stable networks is equal to the weak core of the matching game.*

Interpretation: Chain stability is not the only possible stability notion, but it has strong connections to larger blocking structures in the relevant cases.

5 Tables, figures, and original panels

The paper has no empirical tables. Its visuals are theoretical diagrams and formal equation/theorem panels. The panels below are directly cropped from the paper and grouped compactly.

5.1 Figure 1: unstable and stable networks

Read:

- The example has two upstream suppliers (a_1, a_2), two intermediaries (b_1, b_2), and two final consumers (c_1, c_2).
- Panels A–C show networks that fail chain stability: one is empty, one is blocked by a single contract, and one is not individually rational.
- Panel D shows a chain-stable network.

Economic message: The figure illustrates why blocking coalitions must be chains: profitable deviations in vertical markets propagate through upstream and downstream contracts.

5.2 Figure 2: the T-algorithm

Read:

- The algorithm begins at $\nu_{\min}$, the pre-network with all upstream arrows and no downstream arrows.
- Iterating T adds arrows until a fixed point is reached.
- Applying F keeps only reciprocal arrows and produces the stable network $\mu^*_{\min}$.

Economic message: The algorithm makes the existence proof constructive. It also identifies a particular side-optimal stable network.

5.3 Figure 3: all chain-stable networks in the example

Read:

- The example has four chain-stable networks.

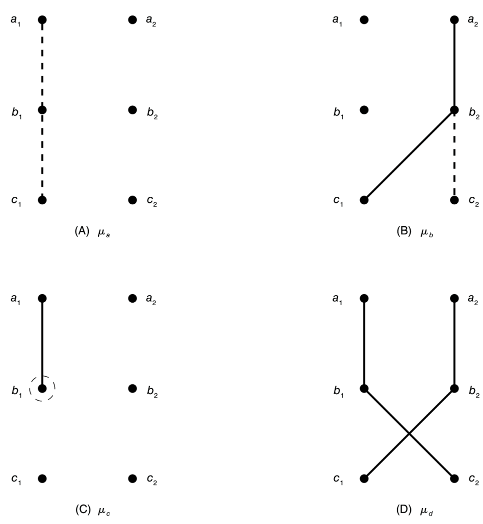

Figure 1. Unstable and Stable Networks

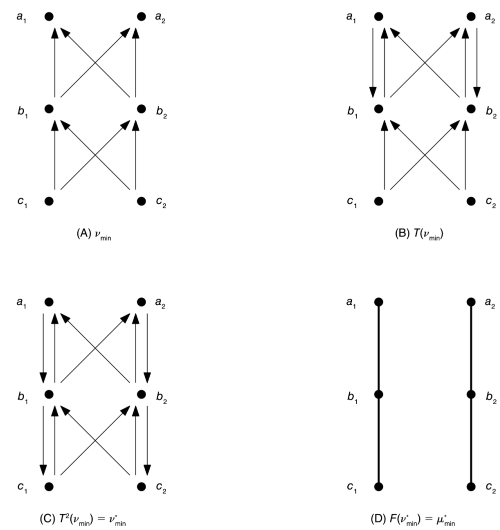

Figure 2. T-Algorithm

- Panel A is the supplier-optimal stable network.
- Panel B is the consumer-optimal stable network.
- Panels C and D are intermediate stable networks.

Economic message: The stable set can contain several networks, but it has well-defined extreme elements for the endpoints of the supply chain.

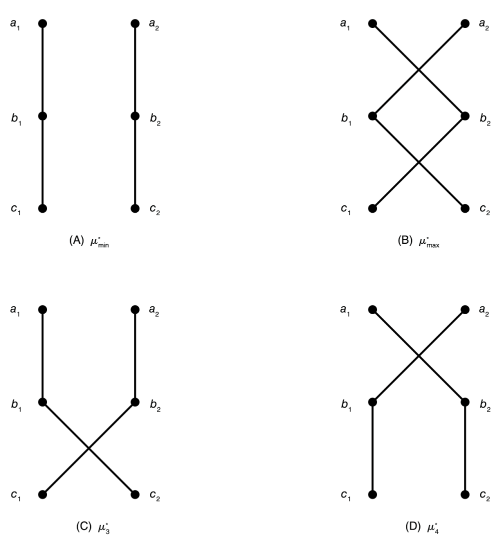

Figure 3. All Chain-Stable Networks

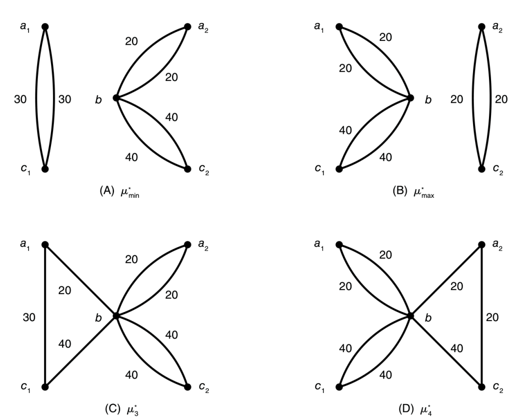

Figure B1. Chain-Stable Networks

5.4 Figure B1: appendix example with prices and quantities

Read:

- Appendix B shows that the framework can handle prices, serial numbers, and multiple units.

- The example includes direct trade and trade through an intermediary.
- The supplier-optimal and consumer-optimal networks differ, and intermediate stable networks also exist.

Economic message: The appendix figure shows that the model is not restricted to simple unit matching without prices. It can represent more realistic supply-chain contracting.

6 Short summary

- The paper generalizes two-sided matching theory to vertical supply-chain networks.
- Agents have fixed upstream–downstream roles.
- Contracts are bilateral and can include prices, quantities, and identities.
- Networks are sets of contracts.
- Chain stability requires individual rationality and absence of blocking upstream–downstream chains.
- The key preference restrictions are same-side substitutability and cross-side complementarity.
- Under these restrictions, chain-stable networks exist.
- The proof constructs a fixed point of the T mapping over pre-networks.
- The T-algorithm starting from $\nu_{\min}$ yields a supplier-optimal chain-stable network.
- The symmetric algorithm starting from $\nu_{\max}$ yields a consumer-optimal chain-stable network.
- Adding suppliers helps consumers and hurts other suppliers; adding consumers has the opposite effect.
- Chain stability coincides with tree stability under the paper's assumptions and with the weak core in a special one-upstream/one-downstream case.
- The framework can be used for market design and as a positive theory for empirical supply-chain analysis.

7 Conclusion

7.1 Core conclusion

Ostrovsky shows that stable matching theory can be extended from two-sided markets to vertical supply-chain networks. Under same-side substitutability and cross-side complementarity, stable networks exist and can be found constructively.

7.2 Economic intuition

The model works because vertical trade has a monotonic structure. Inputs and outputs are complements, but inputs are substitutes among themselves and outputs are substitutes among themselves. This makes the set of unilateral contract proposals move monotonically under the T mapping and creates fixed points that correspond to stable networks.

7.3 Takeaway

The paper provides a foundational theory for matching in supply chains. Its main lesson is that stability in vertical networks requires blocking deviations to respect the supply-chain direction. With the right substitutability and complementarity assumptions, the rich structure of two-sided matching–existence, side-optimality, and comparative statics–survives in much more complex production networks.